

SP-14 Green Chess AI

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

4850-01

Spring 2026

Professor Perry

February 8, 2026

SP-14 Green Chess AI

			
Brett Talley 706-483-9553	Kenneth Burke 925-822-4937	Marc-Henry Moise 786-617-2013	Vladimir Ryzhov 470-775-1900

Table of Contents

1.0	Introduction	3
1.1	Overview	3
1.2	Project Scope	3
1.4	Assumptions.....	3
1.5	References	3
2.0	Functional Requirements	4
2.1	General Requirements	4
2.2	PvP Requirements	4
2.3	PvAI Requirements.....	4
3.0	External Interface Requirements	4
3.1	User Interface Requirements	4

1.0 Introduction

1.1 Overview

Create a self-playing AI Chess Engine that offers both player vs player and player vs

AI game modes. You may use any technology.

Objective:

Create a working chess game which follows the standard rules of chess. The chess has to have two game modes: Player vs Player (PvP) and Player vs Artificial Intelligence (PvAI). In case of abundance of time, we might decide to pursue the objective of adjustable difficulty for the AI game mode.

Code in Python using Pygame Module.

1.2 Project Scope

Working chess, meaning:

- No illegal moves
- En passant, castling

PvP chess

- White cannot move black's pieces and vice versa
- Possibly board will flip every move

PvAI chess

- Engine automatically plays a move when player makes a move

1.4 Assumptions

This will not be an online game; we will be developing the PvP section under the assumption that both players are using the same computer.

1.5 References

<https://www.youtube.com/watch?v=OpL0Gcfn4B4>

- Code game design for board and pieces

<https://www.freecodecamp.org/news/simple-chess-ai-step-by-step-1d55a9266977/>

- Creation of a simple chess ai

<https://chessboardjs.com/>

- Easy embedment of a chess board onto a website

2.0 Functional Requirements

2.1 General Requirements

- Pawn must be able to move forward two squares on first move.
 - o Down the board for black, up the board for white.
 - o Pawns must not be able to move backwards.
 - o Pawn captures must only be available on the immediate diagonal in their respective direction.
 - o En passant captures must be available
- Rooks must be able to move horizontally and vertically along the board.
- Bishops must be restricted to their diagonal.
- Knights must be able to move in offsets of 2 and 1 in an L shape and not be restricted by pieces in their way.
- Queen must be able to have the movement of both the rook and the bishop;
 - o Move along the diagonals and the straight lines.
- King must be able to move one square in any direction.
 - o King must not move into a square where it can be captured.
- Pieces should not be able to move if it exposes the king to be captured
 - o Enforce pins, don't let the king move into an attacked square.
- Castling should be properly implemented.
 - o No castling out of check.
 - o No castling if the squares in between the king and rook are attacked.
 - o No castling if the squares in between have pieces residing on them.

2.2 PvP Requirements

- 2 players
- Can only move their respective pieces

2.3 PvAI Requirements

- Engine automatically generates the best move
- If time allows it, rank the moves
- Ranking the moves allows for adjustable difficulty

3.0 External Interface Requirements

3.1 User Interface Requirements

Mandatory:

- Menu with choice of PvP and PvAI
- Chess pieces graphics

Optional (complete if there is enough time):

- Show legal moves on piece clicked
- Choice of difficulty vs AI
- Option to resign/draw